

Spring Core Maven Exercise 2 | Cognizant DN 5.0

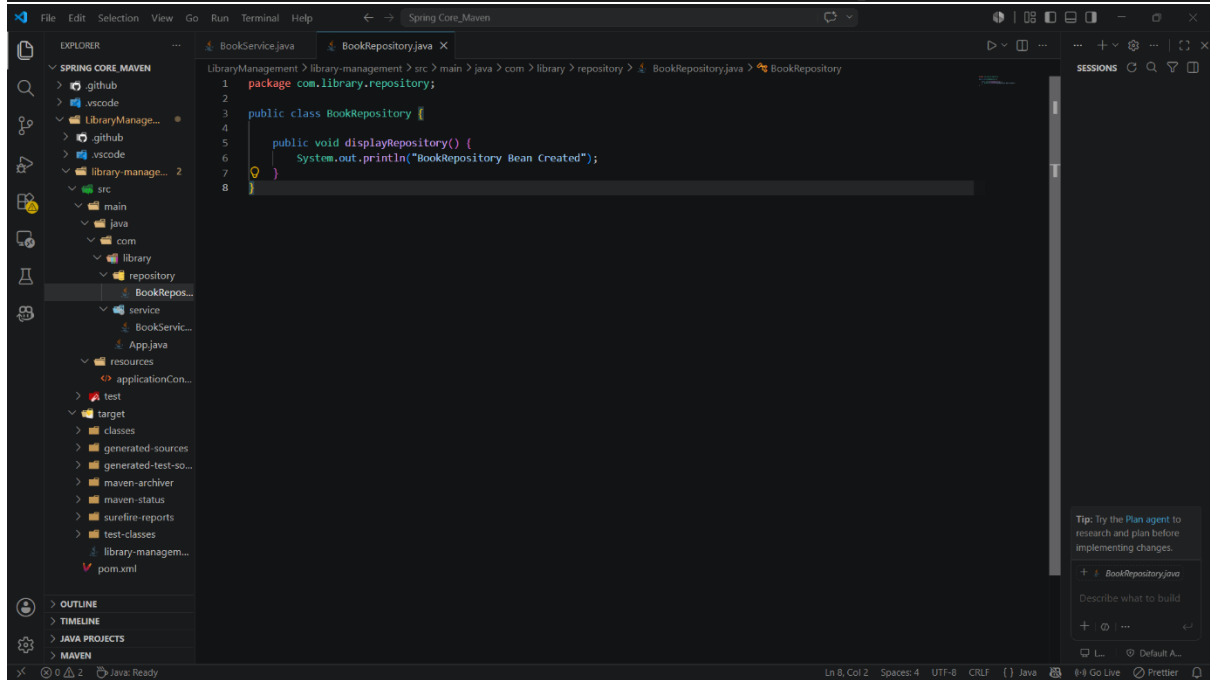
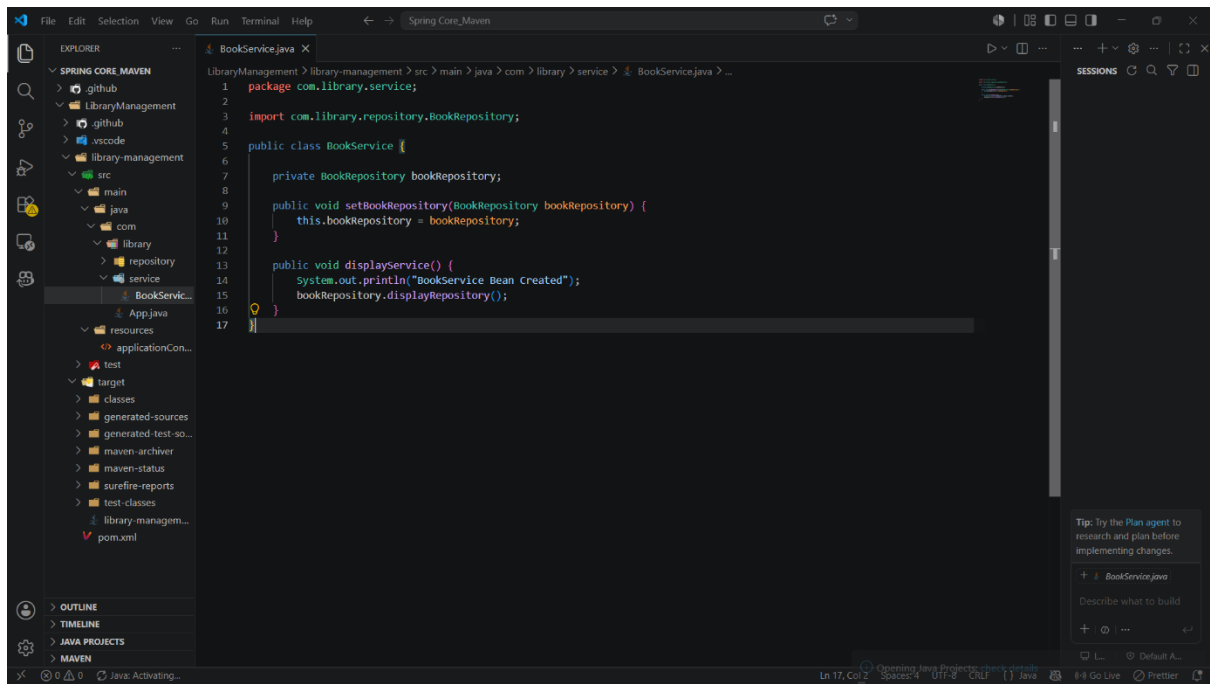
SANSKAR KUMAWAT

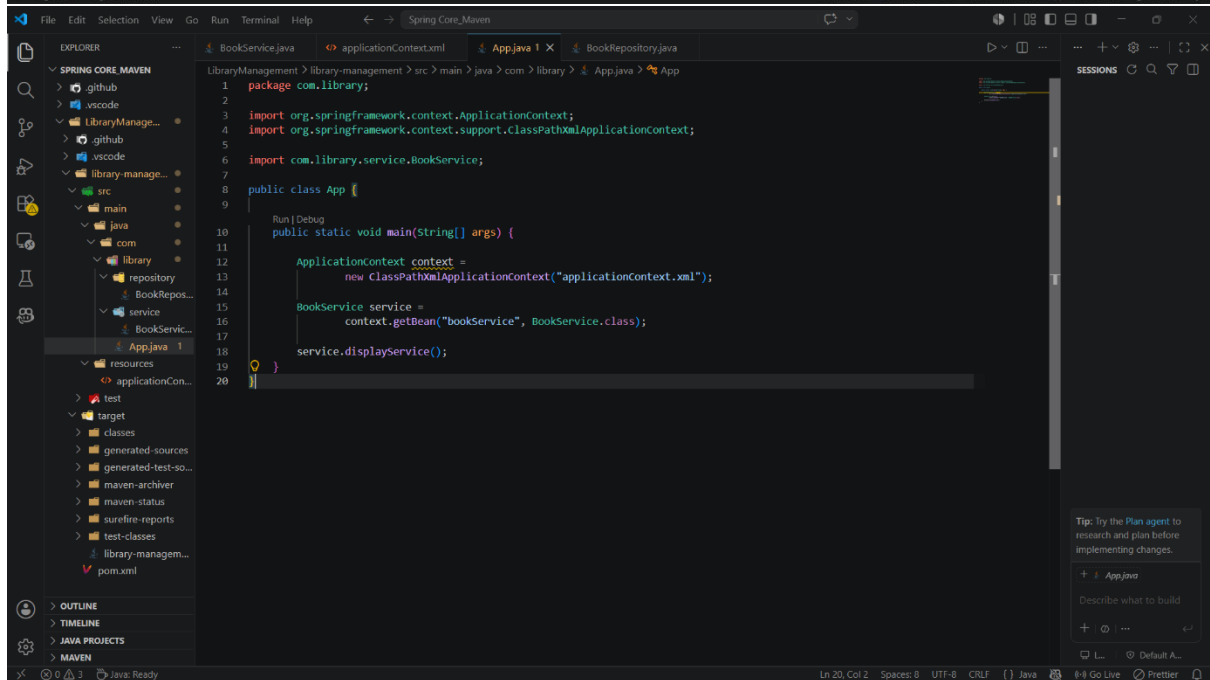
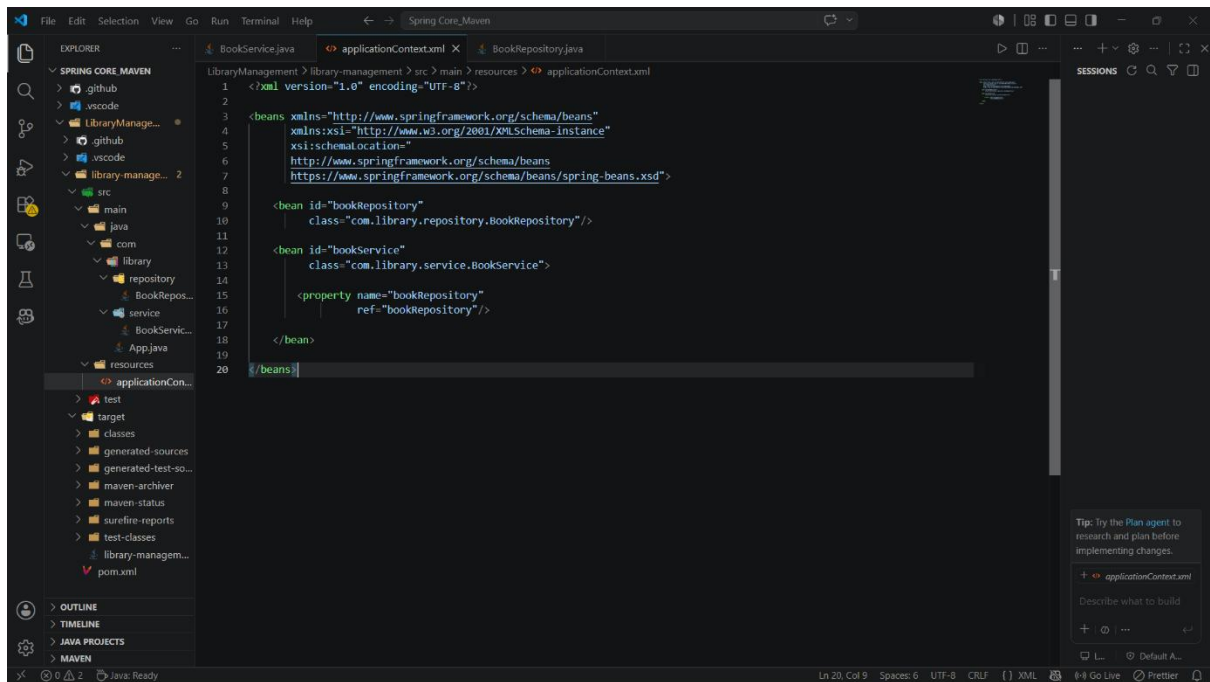
Exercise 2 - Implementing Dependency Injection

Opened the existing **LibraryManagement** Maven project created in Exercise 1.

1. Created the required packages:
 - com.library.service
 - com.library.repository
2. Created the BookRepository class inside the repository package.
3. Created the BookService class inside the service package.
4. Added a private BookRepository object inside the BookService class.
5. Implemented a **setter method** (setBookRepository()) in BookService to enable **Setter Dependency Injection**.
6. Added methods in both classes to display confirmation messages:
 - displayRepository()
 - displayService()
7. Modified the applicationContext.xml file to register both classes as Spring beans.
8. Configured **Dependency Injection** using the <property> tag to inject the BookRepository bean into the BookService bean.
9. Updated the App.java class to:
 - Load the Spring IoC Container using ClassPathXmlApplicationContext.
 - Retrieve the BookService bean using getBean().
 - Invoke the displayService() method.
10. Compiled the project successfully using: mvn clean compile
12. Executed the application using: mvn exec:java "-Dexec.mainClass=com.library.App"
13. Verified that Spring successfully created the beans and injected the dependency.

SCREENSHOTS V





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File Edit Selection View Go Run Terminal Help
Spring Core_Maven

EXPLORER
SPRING CORE MAVEN
  .github
  .vscode
  LibraryManage...
  .github
  .vscode
  library-manage...
    src
    main
      java
        com
          library
            repository
              BookRepos...
            service
              BookServic...
  AppJava 1

resources
  applicationCon...
  test
  target
  classes
  generated-sources
  maven-status
  pom.xml

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
powershell - library-management

PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn clean compile
[INFO] Building library-management 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- clean:3.1.0:clean (default-clean) @ library-management ---
[INFO] Deleting D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management\target
[INFO] --- resources:3.0.2:resources (default-resources) @ library-management ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] Copying 1 resource
[INFO] --- compiler:3.8.0:compile (default-compile) @ library-management ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 3 source files to D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management\target\classes
[INFO] BUILD SUCCESS
[INFO] Total time: 2.104 s
[INFO] Finished at: 2026-07-12T03:59:33+05:30
[INFO] -----
```

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PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn exec:java -Dexec.mainClass=com.library.App
[ERROR]
[ERROR] For more information about the errors and possible solutions, please read the following articles:
[ERROR] [Help 1] http://wiki.apache.org/confluence/display/MAVEN/LifecyclePhaseNotFoundException
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn exec:java "-Dexec.mainClass=com.library.App"
[INFO] Scanning for projects...
[INFO] -----< com.library:library-management >-----
[INFO] Building library-management 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.1.0:java (default-cli) @ library-management ---
BookService Bean Created
BookRepository Bean Created
[INFO] BUILD FAILURE
[INFO] Total time: 0.922 s
[INFO] Finished at: 2026-07-12T04:00:43+05:30
[INFO] -----
```

```
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management> mvn exec:java "-Dexec.mainClass=com.library.App"
[INFO] Scanning for projects...
[INFO] -----< com.library:library-management >-----
[INFO] Building library-management 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] --- exec:3.1.0:java (default-cli) @ library-management ---
BookService Bean Created
BookRepository Bean Created
[INFO] BUILD SUCCESS
[INFO] Total time: 0.922 s
[INFO] Finished at: 2026-07-12T04:00:43+05:30
[INFO] -----
PS D:\VIT\cognizant DN 5.0\Spring Core_Maven\LibraryManagement\library-management>
```